Exclusion of a finite prime-covering period in Erdős Problem #203

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Abstract

Erdős Problem #203 asks whether some positive integer m, coprime to 6, makes every $2^k 3^\ell m + 1$ composite for $k, \ell \geq 0$. We rule out a specific finite-prime-covering approach: no *pure* covering by prime-divisibility cosets has covering period dividing $N = \operatorname{lcm}(1, \ldots, 17) = 12\,252\,240$. A complete computation of the relevant prime pool gives 192 distinct primes. Combining the known slope-one line-projection method with explicitly specified nonnegative weights on residue classes modulo 60 yields the exact upper bound $3\,961\,943/4\,084\,080 < 1$, contradicting the intensity required by a covering. The result extends immediately to every divisor of N by inclusion of prime pools. We provide independent exact-arithmetic verification, including a GMP recomputation of the 2,432-bit pool gcd. A rational linear-programming dual certificate additionally proves optimality of the weights *within this fixed cut and modulus*. The full Erdős problem remains open.

Keywords: covering systems; multiplicative orders; finite cyclic groups; line projection; computational number theory; exact certificates.

1 Problem and precise scope

Erdős and Graham asked whether there is an integer $m \geq 1$ with $\gcd(m, 6) = 1$ such that

$$2^k 3^\ell m + 1 \text{ is composite for every } k, \ell \geq 0. \tag{1}$$

This is Erdős Problem #203 [1, 2]; the unrestricted problem remains open. We consider only the *pure finite prime-covering* route, with no additional algebraic factorisations from special choices of m.

For each prime $p > 3$, define

$$H_p = \langle 2, 3 \rangle \subseteq \mathbb{F}_p^\times, \qquad e_p = |H_p| = \operatorname{lcm}\big(\operatorname{ord}_p(2), \operatorname{ord}_p(3)\big), \tag{2}$$

and the homomorphism $\phi_p : \mathbb{Z}^2 \to H_p$, $(k, \ell) \mapsto 2^k 3^\ell$. Its kernel L_p has index e_p. If p divides some $2^k 3^\ell m + 1$, the set of exponent pairs at which p divides this expression is one coset of L_p. (If $p \mid m$ it divides none.)

A *pure finite prime covering* is a finite family of such cosets, with at most one coset per distinct prime, whose union covers $\mathbb{Z}^2$. Period T means the least common multiple of the indices e_p of the primes used. Replacing $\mathbb{Z}^2$ by nonnegative exponent pairs does not change the covering condition: all the cosets are periodic in both coordinates. This formulation is a necessary condition for a divisibility-only covering construction, not a characterisation of all possible solutions of (1).

Put $N = 12\,252\,240 = \operatorname{lcm}(1, \ldots, 17)$, and

$$\mathcal{P}_N = \{p > 3 \text{ prime} : e_p \mid N\}. \tag{3}$$

Theorem 1. *No pure finite prime covering for Erdős Problem #203 has period dividing $N = 12\,252\,240$.*

The proof combines the slope-one line restriction developed in prior work [3] with a new exact weighted certificate for this larger prime pool. It does not assert that no integer m satisfying (1) exists.

2 A finite line-projection obstruction

The following elementary fact is included for self-containment. The coset-restriction approach is not new; our application uses it in direction $(1,1)$, following [3].

Lemma 1 (Density on a slope-one line)**.** *Let $p>3$ be prime, and set*

$$h_p=\operatorname{ord}_p(6),\qquad i_p=e_p/h_p,\qquad d_p=\gcd(i_p,60).$$

For each coset C_p of L_p, there exists $a_p \pmod{i_p}$ such that, on the diagonal line $k-\ell=\tau$, the proportion of exponent pairs in C_p is

$$\rho_p(\tau)=\begin{cases} i_p/e_p, & \tau\equiv a_p \pmod{i_p},\\ 0, & \text{otherwise.}\end{cases}$$

When these lines are pooled by $\tau \pmod{60}$, the mean proportion contributed by p to class t is

$$\overline{\rho}_p(t)=\begin{cases} d_p/e_p, & t\equiv a_p \pmod{d_p},\\ 0, & \text{otherwise.}\end{cases} \tag{4}$$

Proof. Every point with $k-\ell=\tau$ is of the form $(\tau+u,u)$; hence its image under ϕ_p is $2^\tau 6^u$. The quotient $H_p/\langle 6\rangle$ is cyclic of order i_p, generated by $2\langle 6\rangle$, since $\langle 2,3\rangle=\langle 2,6\rangle$. Membership in a fixed coset of L_p is therefore possible on precisely one class $\tau\equiv a_p \pmod{i_p}$. On such a line, exactly one u-class modulo h_p satisfies it, yielding proportion $1/h_p=i_p/e_p$. Inside a fixed class $t \pmod{60}$, the congruence $\tau\equiv a_p \pmod{i_p}$ has relative density d_p/i_p if $t\equiv a_p \pmod{d_p}$, and 0 otherwise. Multiplying by i_p/e_p proves (4). All densities can equivalently be calculated on the finite torus $(\mathbb{Z}/N\mathbb{Z})^2$, because the relevant orders divide N. □

For $p=5$, $e_5=i_5=4$ and $d_5=4$; for $p=7$, $e_7=6$, $h_7=2$, $i_7=d_7=3$. A single global translation of the exponent lattice moves both selected residues $a_5 \pmod 4$ and $a_7 \pmod 3$ to 0 simultaneously, by the Chinese remainder theorem. We may hence normalise

$$a_5\equiv 0 \pmod 4,\qquad a_7\equiv 0 \pmod 3. \tag{5}$$

This translation does not affect whether the cosets cover the exponent lattice.

Lemma 2 (Weighted line obstruction)**.** *Let $(\lambda_t)_{t\in\mathbb{Z}/60\mathbb{Z}}$ be nonnegative rational weights with $\sum_t \lambda_t=1$, supported on classes t with $t\not\equiv 0 \pmod 3$ and $t\not\equiv 0 \pmod 4$. For each $d\mid 60$ put*

$$\mu_d=\max_{c \pmod d}\sum_{\substack{t \pmod{60}\\ t\equiv c \ (d)}}\lambda_t,\qquad R_d=\sum_{\substack{p\in\mathcal{P}_N\setminus\{5,7\}\\ d_p=d}}\frac{1}{e_p}.$$

If

$$B(\lambda):=\sum_{d\mid 60} d\,\mu_d R_d<1, \tag{6}$$

then the cosets of primes from $\mathcal{P}_N$ cannot cover $\mathbb{Z}^2$.

Proof. Suppose they cover. On each line $k-\ell=\tau$, the sum of coset densities is at least one by the union bound. The same is true after averaging over all lines $\tau\equiv t \pmod{60}$. Take the weighted mean over t using λ_t. The normalised primes $5,7$ contribute zero because the support avoids their classes in (5). By Lemma 1, each other prime p contributes at most $(d_p/e_p)\mu_{d_p}$, whatever residue a_p it selects. Thus the weighted sum is both at least 1 and at most $B(\lambda)$, contradicting (6). □

3 Complete finite prime pool and exact certificate

The crucial finiteness reduction is

$$p \in \mathcal{P}_N \iff p \mid \gcd(2^N - 1, 3^N - 1) =: G_N \qquad (p > 3 \text{ prime}). \tag{7}$$

Indeed $e_p \mid N$ is equivalent to $2^N \equiv 3^N \equiv 1 \pmod{p}$. A provided factorisation into primes is therefore *complete* once the factors are proved prime and their product is checked to equal G_N.

Our data contain 192 distinct primes, the largest being 8 168 161. Exactly $5, 7, 11, 13, 17$ have valuation 2 in G_N; the remaining 187 have valuation 1. The full factor product has **2,432 bits and exactly 732 decimal digits**. The last number is corrected from the initial frozen report, which had used a GMP base-size estimate rather than the exact length of its decimal expansion. An independent GMP computation verified the equality between the product and the directly calculated gcd; a separate Python checker verifies primality and all multiplicative orders.

Choose the explicit weights

$$\lambda_t = \begin{cases} 1/20, & t \in \{1, 26, 41, 46\}, \\ 1/10, & t \in \{2, 5, 7, 10, 14, 19, 23, 58\}, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

They are nonnegative, sum to 1, and vanish whenever $t \equiv 0 \pmod 3$ or $t \equiv 0 \pmod 4$.

For transparency, Table 1 displays the entire grouped exact-rational calculation. The last column is the contribution $N\, d\mu_d R_d$; the relevant nonzero values of d are $1, 2, 3, 4, 5, 6, 10, 12, 15, 20$.

Table 1: The exact weighted obstruction grouped by $d = \gcd(i_p, 60)$ (excluding primes 5 and 7).

d	μ_d	R_d	$Nd\mu_d R_d$
1	1	319967/556920	7,039,274
2	1/2	53617/291720	2,251,914
3	1/2	120079/2042040	1,080,711
4	1/2	91723/4084080	550,338
5	1/5	92789/2450448	463,945
6	1/4	10343/875160	217,203
10	1/10	16/36465	5,376
12	1/4	181/32760	203,082
15	1/10	61/16380	68,442
20	1/10	1/4420	5,544
		Total NB	**11,885,829**

In particular,

$$\boxed{B = \frac{11\,885\,829}{12\,252\,240} = \frac{3\,961\,943}{4\,084\,080} < 1, \qquad 1 - B = \frac{122\,137}{4\,084\,080}.} \tag{9}$$

This and Lemma 2 prove that the full N-pool does not admit a pure covering.

Proof of Theorem 1. If a pure covering has period $T \mid N$, every one of its primes has $e_p \mid T$ and hence lies in $\mathcal{P}_N$. The same nonnegative weights and the same bound for the *full* pool apply, and deleting any primes cannot increase its upper bound. Equation (9) is already below 1. Consequently no such covering can exist. □

The divisor-count refinement in the earlier experimental report is unnecessary for this argument: since $N = 2^4 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, it has $5 \cdot 3 \cdot 2^5 = 480$ positive divisors, all covered at once by monotonicity. This does *not* assert that each of these 480 periods is a previously unknown exclusion; many smaller periods were already treated in [3].

4 Optimality within the fixed weighted cut

Finding the rational weights in (8) used computational optimisation, but the obstruction itself needs no optimiser. Here we can prove an additional, narrowly scoped fact.

Let $S = \{t \pmod{60} : 3 \nmid t,\ 4 \nmid t\}$, so $|S| = 30$. Retain the primes $5, 7$ normalised as above, the slope-one line direction and modulus 60. Over all weights $\lambda_t \geq 0$ supported on S, minimising $B(\lambda)$ is the linear program

$$\begin{aligned} &\text{minimise } \sum_d dR_d z_d, \\ &\text{subject to } \sum_{t\in S} \lambda_t = 1, \quad \lambda_t \geq 0, \\ &\qquad z_d \geq \sum_{t\in S \atop t\equiv c \pmod d} \lambda_t \quad (\text{all } d \text{ with } R_d > 0,\ c \pmod d). \end{aligned} \tag{10}$$

Its dual admits nonnegative rational values $q_{d,c}$ and a lower bound L whenever

$$\sum_{c \pmod d} q_{d,c} \leq dR_d \quad (\forall d), \qquad \sum_d q_{d,t \bmod d} \geq L \quad (\forall t \in S). \tag{11}$$

Indeed, multiplying the first constraints by primal variables or averaging the second ones against λ shows $B(\lambda) \geq L$ for every admissible λ.

Proposition 1 (Optimality for the fixed cut)**.** *For the line direction* $(1, 1)$*, normalised 5–7 support and weights on* $S \pmod{60}$*, the minimum of* $B(\lambda)$ *equals* $3\,961\,943/4\,084\,080$*.*

Proof. The weights (8) give the upper bound in (9). Appendix A supplies an explicit sparse rational assignment $q_{d,c}$ satisfying (11) with $L = 3\,961\,943/4\,084\,080$. Every inequality is checked using rational arithmetic in the companion verifier. Hence weak duality gives the matching lower bound. □

This claim of optimality is *not* a claim of optimality among all possible directions, moduli or prime-covering obstructions.

5 Verification, prior work and limitations

The supplementary verification package contains (i) the complete factor list (with original Git provenance), (ii) a standard-library-only checker of primality, orders, weights and the exact bound, (iii) a GMP/C checker that recomputes G_N from $2^N - 1$ and $3^N - 1$, (iv) rational dual data and a solver-free checker, and (v) machine-readable results. Integer arithmetic and exact fractions, rather than decimal approximations, determine every asserted inequality. The optimisation routine is not needed to rerun the proof.

The 5–7 slope-one line cut, the associated density method and several smaller-period exclusions are due to earlier work recorded by Erdős Problem a Day on 14 October 2026 [3]. The present finite-period computation extends the explicit obstruction to $N = \mathrm{lcm}(1, \ldots, 17)$, which that account identified as a subsequent frontier. A preliminary literature search and independent informal review are encouraging but *not* a proof of exhaustive historical novelty or a substitute for professional mathematical peer review. The mathematical argument above and the included exact verifiers are intended to stand on their own.

The author used an AI-assisted research workflow, including SAPIO MORPHEUS/COMPLEX and ChatGPT, for algorithm design, computation, checking and preparation of this draft.

All mathematical claims are explicitly limited to the certified class and remain the author's responsibility. We claim neither a full solution to Erdős #203 nor any conclusion about non-pure or nonperiodic mechanisms.

References

[1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographie de L'Enseignement Mathématique 28, Université de Genève (1980), p. 27.

[2] T. F. Bloom (ed.), "Erdős Problem #203," https://www.erdosproblems.com/203 (accessed 1 November 2026).

[3] Erdős Problem a Day (Patrick White and Claude), "Erdős #203: along the lines of slope 1 the primes 5 and 7 leave half the lines to the rest of the pool," 14 October 2026, https://erdosproblemaday.com/day/203-the-quotient-lemma-lines-close-the-open-pools.

A An explicit dual certificate

For completeness, the following are the *nonzero* entries $q_{d,c}$ of a feasible dual solution for (11). Unlisted entries vanish. Set $L = 3\,961\,943/4\,084\,080$. The companion file `optional_lp_dual.json` and the Python standard-library script `verify_dual_stdlib.py` check all $\sum_c q_{d,c} \le dR_d$ constraints and all 30 inequalities $\sum_d q_{d,t \bmod d} \ge L$ exactly, with no floating point operations.

d	c	$q_{d,c}$	d	c	$q_{d,c}$
1	0	319967/556920	10	0	161/58344
2	0	16519/157080	10	5	95/58344
2	1	31523/120120	12	10	181/2730
3	1	27697/291720	15	2	61/5460
3	2	83179/1021020	15	5	61/5460
4	2	91723/1021020	15	8	61/5460
5	0	89597/2450448	15	11	61/5460
5	1	7199/188496	15	14	61/5460
5	2	7199/188496	20	2	1/884
5	3	7199/188496	20	6	1/884
5	4	7199/188496	20	14	1/884
6	2	28019/408408	20	18	1/884
6	5	1569/680680			

Thus the proof of Proposition 1 can be checked by addition and comparison of explicit rational numbers. The main theorem does not depend on this proposition.